

The Citation-Faithfulness Illusion: Measuring Whether Grounded Answers Preserve Source Disagreement

Robert Sneiderman

Independent

robbysneiderman@gmail.com

Abstract

Grounding faithfulness is measured on two axes: whether the answer follows the retrieved context, and whether each claim is supported by the source it cites. Both can be satisfied in full by an answer that picks one side of a genuinely contested question, cites a source that holds that side, and drops the credible sources that disagree. We name this failure disagreement collapse, or the citation-faithfulness illusion, and treat it as a construct-validity hazard: a high citation score does not license the inference that the answer reflects the sources. Conflict-aware retrieval already prescribes reflecting credible debate; our contribution is to make that prescription measurable. We define disagreement recall and disagreement-attribution precision, both decoupled from answer correctness and conditioned on source credibility so that suppressing a debunked claim is not penalized; a five-component rubric with verbatim anchors; an annotation protocol for the credibility filter and for position identity; a matched-pair diagnostic that holds retrieval and citation availability fixed; and a pre-registered two-judge pilot with human adjudication. No empirical scores are reported; the contribution is the instrument and its frozen evaluation design.

1 Introduction

A grounded answer can be perfectly attributed and still misrepresent the evidence. Consider a contested factual question on which three credible retrieved sources hold two distinct positions. An answer adopts the majority position, cites the one source that supports it with full citation precision and recall, presents the matter as settled, and silently drops the credible sources on the other side. The widely used entailment and attribution metrics are all satisfied. Entailment checkers find the answer entailed by its context; citation checkers find each claim supported by its cited source.

The answer reads as well-grounded and trustworthy. What it conceals is that the cited corpus disagrees. We call this failure *disagreement collapse*, and the metric pattern that makes it invisible the *citation-faithfulness illusion*: grounding can make an answer look more trustworthy while making it less faithful to the actual state of the evidence.

This is a third faithfulness axis, and it is currently unmeasured. Entailment metrics ask whether the answer follows the retrieved context (Es et al., 2024; Honovich et al., 2022; Tamber et al., 2025). Attribution metrics ask whether each claim is supported by its cited source (Rashkin et al., 2023; Gao et al., 2023; Bohnet et al., 2022; Qi et al., 2024; Liu et al., 2023).

Neither asks whether the sources agree, so neither can detect an answer that resolves a live disagreement into one confident, correctly-attributed claim. The axis we name, disagreement-preservation faithfulness, asks a different question: given retrieved sources that genuinely conflict on a contested factual question, does the grounded answer represent the existence and structure of that disagreement? The unit of disagreement is source-versus-source, not query ambiguity: AmbigDocs (Lee et al., 2024) and ASQA (Stelmakh et al., 2022) ask an answer to cover several entity-level readings of one ambiguous query, whereas the question here is whether credible sources hold conflicting positions on one contested fact and whether the answer surfaces them.

The failure is a response-layer instance of consensus collapse, the compression of meaningful disagreement into a single output whose presentation conceals what was compressed, and it is a construct-validity hazard in the sense of Jacobs and Wallach (2021): a high citation-precision score does not license the inference that the answer reflects the sources, because the score measures claim-to-source support and the inference concerns fidelity to the source distribution. The

two are not the same construct, and one can be maximal while the other is zero. Alignment by RLHF and DPO pushes output toward majority views (Wright et al., 2025), so a grounded model confidently picking one side of a contested question is the expected behavior under current objectives, not an edge case.

The desideratum is narrow. It applies only to *legitimately contested* questions, and the metrics are conditioned on source credibility, so suppressing a debunked claim is correct boundary handling rather than penalized collapse, which pre-empts the objection that preserving disagreement amounts to both-sidesing misinformation. The credibility filter does the work here, and we annotate it per source rather than assume an oracle. The motivating analogue is interpretive agency: a reader’s ability to see and check the basis of a grounded claim, which frames faithfulness as a transparency-of-basis property rather than an aggregate quality score.

The contribution is narrow.

1. A named, orthogonal faithfulness axis, disagreement-preservation faithfulness, separated by a constructive argument from entailment-faithfulness and attribution-faithfulness and framed as a construct-validity failure (Jacobs and Wallach, 2021).
2. A metric pair decoupled from answer correctness, disagreement recall and disagreement-attribution precision, conditioned on source credibility so that suppressing a debunked claim is not penalized as collapse.
3. A verbatim five-component response-level rubric (Visibility, Tradeoffs, Conflict Calibration, Credibility Boundaries, Actionability) with one-to-five paragraph anchors.
4. The citation-faithfulness illusion exhibited as a worked, firing artifact on one constructed item: two answers indistinguishable under citation precision and entailment-faithfulness by construction are separated by the primitive, with a typed item-level trace.
5. A controlled diagnostic design: contested questions paired into matched agree/conflict source variants with retrieval and citation availability held fixed, plus a debunked-claim arm that separates correct boundary handling from collapse, and an annotation protocol for the credibility filter and for position identity.

6. A pre-registered pilot design with no results: multiple production models under one fixed RAG system prompt, two independent LLM judges plus human adjudication on a subset, with per-item gold positions, Krippendorff’s α (Krippendorff, 2004; Landis and Koch, 1977), a pre-declared discriminant-validity rule (Fornell and Larcker, 1981), an inter-dimension matrix, and a same-family judge-bias diagnostic, committed at a fixed hash with bad-results-are-primary commitments.

The boundaries are equally firm. The framework adds no new RAG system and no conflict-resolution method. It is not a general-purpose benchmark; the diagnostic is a small curated seed set with a documented construction protocol, not a leaderboard. Models do not always collapse disagreement, and this submission reports no measured scores. The reliability and discriminant validity of the rubric are hypotheses the pre-registered pilot tests, and the headline negative result is registered in advance, not observed.

2 Two Faithfulness Axes and the Seam Between Them

Grounding faithfulness is operationalized today along two axes. Entailment-faithfulness asks whether the answer follows from the retrieved context, treated as one undifferentiated block (Es et al., 2024; Honovich et al., 2022; Tamber et al., 2025); an answer that adopts one credible position and ignores an equally credible contradicting passage in the same context stays faithful by this axis. Attribution-faithfulness asks whether each claim is supported by the source it cites (Rashkin et al., 2023; Gao et al., 2023; Bohnet et al., 2022; Liu et al., 2023; Qi et al., 2024);

none of these compares cited sources to one another, so an answer can cite a real source for its chosen side with perfect citation precision while dropping the sources that contradict it. Section 8.1 places the individual systems.

The third axis is not derivable from the first two. Consider, schematically, a contested question with a cited source set in which two credible positions are present. Construct an answer that states only the majority position and cites only a source that holds it. The answer is maximally entailment-faithful: every claim it makes is supported by a passage in context. It is maximally

attribution-faithful: every claim it makes is supported by the source it cites, so citation precision and citation recall are both one. And its disagreement recall is one-half, because it surfaces one of two credible positions and drops the other. Entailment and attribution are therefore jointly satisfiable at their maxima while disagreement recall is held down by the dropped credible position, which is the formal sense in which the third axis is orthogonal. The metric pair of Section 3 makes it quantitative.

3 Defining Disagreement-Preservation Faithfulness

We define the metric pair on a fixed contested question and its cited source set, decoupled from whether the answer picks the correct side.

Setup. Let q be a contested factual question and $S = \{s_1, \dots, s_n\}$ the set of sources cited by, or available to, a grounded answer y . Each source s_i carries a position $p(s_i)$ on q and a credibility label $c(s_i)$. A *credibility filter* partitions S into a credible subset $S^+ = \{s_i : c(s_i) = \text{credible}\}$ and a non-credible remainder S^- (debunked, retracted, or noise). The filter is annotated per source in the diagnostic (Section 6) rather than assumed from an oracle, and it is load-bearing: every quantity below conditions on it.

Gold positions. The gold set of distinct credible conflicting positions is

$$P(q, S) = \{p(s_i) : s_i \in S^+\} / \sim,$$

where $\sim$ merges paraphrases of the same position so that two credible sources asserting the same claim count once. A *position* is the stance taken on q , not the source, and merging under $\sim$ is what makes “distinct credible position” the unit of recall rather than “cited source.” Because both $\sim$ and the partition are annotations, $P(q, S)$ is fixed as a per-item gold label, not inferred at scoring time; the pilot (Section 7) reports agreement on enumerating $P(q, S)$ separately from agreement on detection.

Disagreement recall. Let $\text{surf}(y) \subseteq P(q, S)$ be the gold positions that the answer *surfaces*. A position is surfaced when the answer asserts it as a stance held on q and attributes it to at least one source, which is the operational line that separates surfacing from a bare hedge such as “some debate

exists”; the decision procedure is given in Section 4. Then

$$\text{DR}(y) = \frac{|\text{surf}(y)|}{|P(q, S)|}.$$

Disagreement recall is one when the answer surfaces every distinct credible position and zero when it surfaces only one of several. It does not reward surfacing positions drawn from S^- ; a position grounded only in a non-credible source is not in $P(q, S)$ and contributes to neither numerator nor denominator.

Disagreement-attribution precision. For each surfaced position, the answer should tie it to a source that actually holds it. Let $\text{att}(y)$ be the surfaced positions for which the answer’s attribution names a credible source whose position is that position. When $\text{surf}(y) = \emptyset$ we set $\text{DAP}(y) = 0$ by convention, because a fully collapsing answer has nothing correctly attributed. Otherwise

$$\text{DAP}(y) = \frac{|\text{att}(y)|}{|\text{surf}(y)|}.$$

Disagreement-attribution precision penalizes surfacing a position while misattributing it, including the case where an answer lists “some sources say X” without tying X to a source that holds X. It is the attribution analogue at the level of positions rather than claims, and it is exactly the check that ordinary citation precision (Section 2) cannot perform, because ordinary citation precision asks only whether the cited source supports the single claim the answer made.

Decoupling from correctness. Neither metric references which position on q is correct: an answer that surfaces and correctly attributes both sides of a genuine open dispute scores one on both. Suppressing a debunked claim moves neither metric, because S^- is excluded from $P(q, S)$; correct boundary handling is scored under Credibility Boundaries (Section 4), not as collapse.

4 The Source-Conflict Rubric (V/T/C/B/A)

The metric pair scores collapse quantitatively. The rubric is its qualitative companion: a response-level instrument that two independent judges apply to a real answer with paragraph-level anchors, so that the construct has an auditable definition before any data is collected. Each component is

scored on a 1–5 scale. The five components, abbreviated V/T/C/B/A, are Visibility, Tradeoffs, Conflict Calibration, Credibility Boundaries, and Actionability. The anchors below are printed verbatim and are the same text the pilot judges (Section 7) use.

Surfacing: the decision procedure. Because Visibility and disagreement recall both turn on whether a position is “surfaced,” we fix the test once. A position counts as surfaced if and only if the answer both asserts the stance as one held on *q*, in words a reader would identify as a claim rather than a disclaimer, and attaches it to at least one named source. A generic uncertainty hedge that names no position fails the first clause; a position mentioned with no source fails the second and is scored under Tradeoffs and Actionability, not Visibility. Judges apply this two-part test before scoring any facet.

Visibility. *Does the answer surface the distinct credible positions present in the cited sources, rather than presenting one position as settled?* (1) States one position as fact; no signal that credible sources differ. (2) States one position; a hedge (“some debate exists”) without naming a second position. (3) Names a second position but subordinates it as a minor caveat. (4) Names the distinct credible positions as positions, with mild asymmetry in framing. (5) Names every distinct credible position as a position held on the question, without overstating or understating their relative standing. Visibility is the response-level analogue of disagreement recall; the two overlap by design and the pilot’s inter-dimension matrix (Section 7) measures how much.

Tradeoffs. *When more than one position is surfaced, does the answer represent why the sources differ?* (1) No second position to explain. (2) Asserts a bare winner (“studies are mixed, but X”). (3) Notes that sources differ without a reason. (4) Gives a partial basis for the difference (one of methodology, recency, population, scope). (5) Ties the disagreement to its basis, so the reader can see that, for example, two cohort studies diverge because they use different exposure windows.

Conflict Calibration. *Is the kind of disagreement identified correctly, and is each position’s confidence tied to its source standing?* The types are a genuine open dispute among credible sources, a stale-versus-current split where recency

settles it, a debunked-versus-credible split where credibility settles it, and an apparent conflict that reconciles once scope is fixed. (1) Mistypes the conflict in a way that inverts the right handling (an open dispute called settled, or a settled question called open). (2) Mistypes in a milder way. (3) Leaves the type implicit. (4) Types the conflict correctly with minor imprecision. (5) Names the conflict type and ties each position’s stated confidence to its source standing.

Credibility Boundaries. *Does the answer avoid presenting a non-credible or debunked claim as legitimate disagreement?* Visibility pushes toward surfacing positions; Credibility Boundaries is the countervailing guard that keeps that pressure from manufacturing false balance. (1) Elevates a debunked or retracted claim to “one side of an active debate.” (2) Lists a non-credible position as a neutral alternative. (3) Mentions a non-credible position with a weak disclaimer. (4) Marks a non-credible position as such while still naming it. (5) Surfaces only credible positions, and where a non-credible claim is relevant, states that a credible body of evidence has settled it. An answer with high source coverage but elevated misinformation fails Credibility Boundaries even at high Visibility; the rubric scores the two separately so a judge can distinguish a two-sided question handled well from a settled question wrongly presented as two-sided.

Actionability. *Can the reader see what was compressed and check it?* (1) States a recommendation with no basis the reader can inspect. (2) Cites a source but not per position. (3) Ties some positions to inspectable sources. (4) Ties each surfaced position to a source and gives a partial account of what would change the answer. (5) Ties each surfaced position to a source the reader can inspect and states what would change the answer (newer evidence, a different population, a resolved methodological dispute). This is interpretive agency in the grounded setting, faithfulness as a transparency-of-basis property the reader can verify.

Deliberate redundancy. The five-component vector is not assumed orthogonal: Visibility overlaps disagreement recall, and Visibility and Credibility Boundaries pull in opposite directions. Rather than assert separability, the framework commits in advance to measuring the inter-

component correlation structure empirically (Section 7), since a construct defended only by assertion is the construct-validity hazard the framework is built to expose (Jacobs and Wallach, 2021).

5 Firing the Primitive: A Worked Trace

The illusion is exhibited on one constructed, clearly labeled illustrative item. Every value in this section is a property of that construction, never a model result. The item is a contested factual question with three credible cited sources, s_1 and s_2 holding position p_a and s_3 holding position p_b , so $P(q, S) = \{p_a, p_b\}$ with $|P(q, S)| = 2$.

Answer A1 (collapsing). A1 states p_a as the answer, cites s_1 , and presents the question as settled. Under ordinary attribution A1 is exemplary: its single claim is supported by s_1 , so citation precision and recall are both one, and it is supported by the context. The primitive *rejects* A1 with a typed trace: $\text{surf}(A1) = \{p_a\}$, so $\text{DR} = 1/2$ and $\text{DAP} = 1$; the dropped credible position p_b (held by s_3) is named as the collapse; Visibility 1, Tradeoffs 1, Conflict Calibration 1 (an open dispute called settled), Credibility Boundaries 5, Actionability 1.

Answer A2 (preserving). A2 surfaces both p_a and p_b , attributes p_a to s_1, s_2 and p_b to s_3 , types the conflict as a genuine open dispute, and states what would change the answer. The primitive *certifies* A2: $\text{surf}(A2) = \{p_a, p_b\}$, so $\text{DR} = 1$ and $\text{DAP} = 1$; the four content facets score high and Credibility Boundaries holds.

The separation. A1 and A2 are indistinguishable to citation precision and to entailment-faithfulness: both make only claims supported by a cited source, and both are supported by context. The primitive is exactly what separates them, the citation-faithfulness illusion made concrete. The construction is illustrative, not evidence that any model produces it.

6 A Controlled Diagnostic That Isolates Collapse

The diagnostic isolates disagreement collapse as a dependent variable by holding retrieval and attribution fixed and varying only whether the credible sources agree. It is a small curated seed set with a documented construction protocol, a design target rather than collected data.

Matched pairs. Each item is a contested question drawn from a domain with documented legitimate disagreement among credible sources, screened in advance to be low-harm. Every question appears in two matched variants: an agree-variant whose credible sources concur, and a conflict-variant with the same source count, comparable length, identical citation availability, and annotated credibility tiers, but diverging credible positions. Holding retrieval and citation availability fixed leaves source agreement as the only manipulated factor, so collapse is the dependent property the rubric is meant to detect. On the conflict-variant a model can attain full citation precision and recall while collapsing disagreement recall toward its floor, instantiating the illusion of Section 5.

Gold annotations. Each conflict-variant ships a gold $P(q, S)$, a gold credibility tier per source, and the gold position-to-source map, fixed before any model is run. Computing disagreement recall against a frozen gold set, rather than a judge’s own enumeration, is what makes the metric reproducible and what lets the pilot separate disagreement about enumerating positions from disagreement about whether a given answer surfaced them.

Annotation protocol for credibility and position identity. Both partitions are annotated before any model runs, by two annotators with a third adjudicating disagreements, and agreement on each is reported separately from judge agreement. The credibility tier follows a fixed source-type rule: peer-reviewed studies, systematic reviews, and current guidelines from a recognized professional or public body are *credible*; a source is *non-credible* when it has been retracted or when its central claim has been rejected by a credible body of evidence named in the item’s scope note; everything else is *noise* and enters neither S^+ nor the debunked-claim arm. Position identity is decided against the item’s scope statement, a one-sentence fixing of the question’s population, exposure, and outcome: two sources hold the same position when they give the same directional answer under that statement, and different positions when they differ in direction or in a qualifier the scope statement makes material. In the constructed item of Section 5, s_1 and s_2 merge under $\sim$ because both answer q in the direction p_a and differ only in effect size, while s_3 answers in the opposite direction and forms p_b ; all three are

credible by the source-type rule, so S^- is empty and $|P(q, S)| = 2$. Items on which the adjudicator cannot fix a scope statement are dropped and the count of dropped items is reported. Annotation cost is the binding constraint on scale: each item needs a scope statement, per-source tiers, and a position-to-source map, which we estimate at twenty to forty minutes of a domain-literate annotator’s time, so the seed set is small by design and the protocol, not the set, is the reusable artifact.

Construction controls and the debunked-claim arm.

Position balance is matched alongside the controls above so they cannot confound the agree-versus-conflict contrast. A debunked-claim arm augments the conflict-variant with a non-credible source, separating correct credibility-boundary handling from collapse and exercising the Credibility Boundaries facet directly: an answer that suppresses the non-credible source scores well on that facet and is not penalized on the metric pair, because that source is excluded from $P(q, S)$.

Size, licensing, anonymization. Sources are drawn under terms permitting redistribution of the seed set, and the released artifacts carry no deanonymizing information.

7 Pre-Registered Pilot: Reliability and Discriminant Validity

This section reports a protocol, not findings. Design, model set, rater set, and analyses were fixed in a versioned pre-registration committed before any response was generated or scored; the contribution here is the frozen design and its advance commitments, and every quantity below is a design parameter rather than a measured outcome. The pilot answers two questions about the instrument from Sections 3 and 4, both prerequisites for any later claim about model behavior: do independent judges agree when they apply the disagreement-faithfulness metrics and the V/T/C/B/A rubric, and is the resulting construct distinct from a generic answer-quality score rather than a relabeling of it (Jacobs and Wallach, 2021).

7.1 Generation condition

Each pilot item uses a source pack from the controlled diagnostic of Section 6, with its frozen gold $P(q, S)$ and debunked-claim arm. Multiple production models generate one response per item under a single fixed ordinary-RAG system prompt,

fixed temperature, and fixed token budget shared across all models. One generation condition removes prompt tuning as a confound and keeps the pilot a test of the instrument rather than a leaderboard. The prompt instructs each model to answer using the provided sources and to cite them, with no instruction to surface or to resolve disagreement, so the setting reflects ordinary grounded-answering rather than one primed toward the property under measurement. Decoding parameters, prompt text, and model identifiers are fixed in the pre-registration.

7.2 Judging condition

Two independent LLM judges from different model families score every response, and a human adjudicator scores a stratified subset (Section 7.3). Each response is identity-scrubbed and presented in a fixed-seed shuffled order, so a judge cannot condition on the generator or on adjacency between an agree-variant and its conflict-variant. A judge scores each response on disagreement recall, disagreement-attribution precision against the gold $P(q, S)$, the five rubric facets,

and a separate generic answer-quality control H , using the verbatim anchors and the surfacing test of Section 4. The control H is the discriminant-validity yardstick: it captures fluency, helpfulness, and apparent groundedness without reference to disagreement, the signal entailment and citation metrics already reward.

7.3 Pre-declared primary analyses

Three analyses are designated primary in the pre-registration, together with a human anchor and one diagnostic.

(A1) Inter-rater reliability. Krippendorff’s α with ordinal weights is computed per dimension across the two judges, reported as a point estimate with a bootstrap 95% confidence interval (Krippendorff, 2004). The band transfer is heuristic: we read the point estimate against fixed Landis-Koch bands declared in advance (Landis and Koch, 1977), noting that those bands were derived for categorical κ and are used here only as a coarse reference. Bands and weighting are fixed before scoring so the reliability verdict is not chosen after seeing the numbers.

(A2) Human anchor. A human adjudicator scores a stratified subset spanning agree, conflict, and debunked-claim items on every dimension.

Human-versus-each-judge α on this subset is reported alongside judge-versus-judge α . A high judge-judge α with a low judge-human α on the same items is the registered signature of a shared judge blind spot rather than of reliability, and is reported as such.

(A3) Discriminant validity. The disagreement composite, the mean of the five rubric facets together with the two metrics, is correlated with the generic-quality control H using Spearman’s ρ , per judge and rater-averaged. The pre-set rule follows Fornell–Larcker logic (Fornell and Larcker, 1981): the construct counts as distinct only if the squared cross-correlation with H is below the construct’s own reliability, operationalized as $\rho^2 < \alpha_{\text{composite}}$ and, as a fixed floor, $\rho < 0.70$. A round liberal threshold such as $\rho < 0.85$ would admit roughly seventy percent shared variance and is rejected in advance for that reason. Fixing the rule before scoring prevents a post-hoc redefinition of what counts as distinct.

(A4) Inter-dimension structure. The full Spearman correlation matrix across the rubric facets and the two metrics is reported on rater-averaged scores. A pre-declared rule governs the expected Visibility–disagreement–recall redundancy (Section 4): a facet pair with $\rho \geq 0.90$ is flagged for merge in any future revision, pairs below that retained as carrying independent signal.

Same-family judge-bias diagnostic. Because each judge shares a model family with one generator, per-(judge, generated-model) mean composite scores are tabulated to expose any tendency of a judge to favor same-family output, reported alongside the primary analyses so a family effect, if present, is visible rather than buried.

7.4 Advance honest-reporting commitments

Three reporting commitments are made before data exist, each chosen because the corresponding outcome would be tempting to soften after the fact. Any dimension whose α falls in the poor or fair band, or whose judge-human α is materially below its judge-judge α , is reported as a primary finding about the instrument, not a caveat. A composite-versus- H correlation that fails the (A3) rule is reported as a primary finding that the rubric is not yet distinct from generic quality, even though that outcome undercuts the paper’s central

separation claim. No post-hoc analysis is reported as primary; exploratory analyses, should any be run, are labeled as such and kept distinct from the pre-declared set.

7.5 Pre-registration discipline

The full design, all identifiers, the human-anchor protocol, the generation and judging conditions, the four primary analyses, the same-family diagnostic, the interpretation bands, and the discriminant rule are recorded in a versioned pre-registration anchored at a fixed commit hash, committed before any response was generated. Runtime deviations are logged separately rather than edited in. The pilot is cost-capped and sized for a small curated set, two judges, and a human-anchor subset; because bootstrap intervals on a small set may straddle adjacent bands, the pre-registration states a minimum item count below which a banded verdict is reported as indeterminate. The pre-registration, deviations log, verbatim anchors, gold annotations, and analysis code are released so a reader can regenerate the analysis. The headline outcome, that grounded answers with strong citation scores can still collapse genuine source disagreement, is a hypothesis the frozen design tests, not a result reported here.

8 Related Work and Positioning

8.1 Entailment and attribution faithfulness

Entailment-faithfulness, treating the retrieved context as one undifferentiated block, includes RAGAS (Es et al., 2024), TRUE (Honovich et al., 2022), and the FaithJudge LLM-as-judge leaderboard (Tamber et al., 2025). Attribution-faithfulness, judging whether each claim is supported by its cited source, includes the Attributable-to-Identified-Sources framework (Rashkin et al., 2023); ALCE (Gao et al., 2023), which operationalizes

citation recall and precision and reports that on long-form ELI5 even the best models lack complete citation support half of the time; AutoAIS (Bohnet et al., 2022) and verifiability evaluation of generative search engines (Liu et al., 2023); and the model-internals attribution of Qi et al. (2024). None compares cited sources to one another, so a model can cite a real source that supports its chosen side with perfect citation precision while silently dropping the equally credible cited sources that contradict it.

8.2 Knowledge and source conflict: resolution and reflection

A growing line studies what happens when retrieved sources contradict each other. The knowledge-conflict survey (Xu et al., 2024) taxonomizes context-memory, inter-context, and intra-memory conflict. DRAGged into Conflicts (Cattan et al., 2025) detects conflicting sources in search-augmented models

and prescribes a desired behavior per conflict type, including, for genuinely conflicting credible opinions, explicitly reflecting the debate and neutrally summarizing the viewpoints, and, for misinformation, disregarding the non-credible source. RAMDocs and MADAM-RAG (Wang et al., 2025) jointly stress ambiguity, misinformation, and noise, using debate between several model instances, and expect multiple valid answers for ambiguity while suppressing misinformation. These are the nearest neighbors, and our target behavior on the conflict-variant coincides with the behavior these works prescribe. The delta is not the desideratum but how it is made measurable: a quantitative metric pair where these works supply a per-type behavioral prescription scored by an LLM-judge quality signal, the construct-validity reading that a high citation score does not license a fidelity inference, and a controlled matched-pair diagnostic at the unit of source-held positions rather than query interpretations or per-claim support. A head-to-head comparison would score the same conflict items under their per-type judge prescription and under disagreement recall against a frozen gold set; that comparison belongs to the pilot’s exploratory set and is not a result here.

8.3 Diversity, knowledge collapse, and query ambiguity

A fourth line concerns breadth rather than fidelity to source conflict. Epistemic diversity and knowledge collapse (Wright et al., 2025) documents that alignment pushes outputs toward majority views, so collapse of source disagreement is a predicted consequence of that pressure. AmbigDocs (Lee et al., 2024) and ASQA (Stelmakh et al., 2022) target *query* ambiguity, where one question admits several interpretations, not source-versus-source disagreement about one contested fact; DiverseSumm (Huang et al., 2024) measures how much distinct information a summary captures,

not whether it preserves a contradiction. Coverage rewards breadth; disagreement-preservation faithfulness rewards fidelity to the distribution of credible positions, conditioned so breadth toward a debunked claim is never credited.

9 From Measurement to Mitigation

The framework localizes collapse and does not repair it, but the metric pair is shaped to be usable as a target, and three hooks follow from its definition. At the prompt, an instruction to state each credible position with its source before answering raises disagreement recall directly, and the risk it introduces, false balance, is what the Credibility Boundaries facet scores, so a prompt change can be evaluated on both at once. At retrieval, conflict-aware retrievers that group evidence by position (Cattan et al., 2025; Wang et al., 2025) hand the generator the partition

the metric assumes, and the matched-pair diagnostic can separate collapse that happens in retrieval from collapse that happens in generation by holding the positions fixed. In training, the product of disagreement recall and disagreement-attribution precision, gated by a credibility-boundary check, is a

reward that credits surfacing credible positions and never credits surfacing debunked ones; whether such a reward survives optimization without drifting into both-sidesing is a question the pilot’s instrument would be needed to answer. None of the three is run here.

10 Conclusion

Both established faithfulness axes can be satisfied in full by an answer that adopts one side of a contested question and silently drops the conflicting cited sources. We name that failure the citation-faithfulness illusion, frame it as a construct-validity hazard (Jacobs and Wallach, 2021), and make the conflict-aware desideratum (Cattan et al., 2025; Wang et al., 2025) measurable with a credibility-conditioned

metric pair, a verbatim rubric, a worked firing item, a matched-pair diagnostic that holds retrieval and citation availability fixed, and a pre-registered two-judge pilot.

11 Limitations

Three boundary conditions bound what the construct measures.

Scope: legitimately contested questions only.

The desideratum is defined over questions on which credible sources genuinely diverge, and every metric conditions on a per-source credibility filter, so the metric pair is not a both-sidesing objective and does not reward false balance. The cost is that the credibility partition is itself contestable: deciding which positions are credible-conflicting and which are noise is a judgement, and where credibility is disputed the boundary between preservation and amplification is where reasonable annotators may differ. The diagnostic annotates tiers per source rather than assuming an oracle, and the metric is only as defensible as those annotations.

Judges can collapse disagreement too. The same alignment pressures argued to drive collapse in generators (Wright et al., 2025) apply to judges: a judge that internalized a majority view may under-credit a surfaced minority position. The two-family design, the same-family bias diagnostic, and the judge-versus-human anchor partly address this by exposing the case where two judges agree while diverging from a human, but two judges sharing a prior can agree while both collapse, and a single human anchor carries its own prior. Inter-rater reliability bounds noise, not shared bias, so a high α does not on its own certify faithful reading; broader human adjudication is left to the run.

A diagnostic that localizes, not a method that fixes. The contribution is measurement. A low disagreement-recall score localizes a transparency gap but does not repair the generator; nothing here trains or prompts a model toward preservation, and Section 9 names the hooks without running them. The metric also says where collapse occurs, not why: a low score is consistent with the model failing to detect the conflict, resolving it silently, or surfacing it in a register the judge does not credit, and separating those mechanisms needs interventions this design does not run. Construct validity beyond this submission rests on the pilot, whose figures are pre-registered hypotheses; the worked item in Section 5 is a labelled illustrative construction, not evidence of any model’s behaviour. No empirical scores are reported, and the headline negative result is a pre-committed prediction the run will test.

References

- Bernd Bohnet, Vinh Q. Tran, Pat Verga, Roei Aharoni, Daniel Andor, Livio Baldini Soares, Massimiliano Ciaramita, Jacob Eisenstein, Kuzman Ganchev, Jonathan Herzig, Kai Hui, Tom Kwiatkowski, Ji Ma, Jianmo Ni, Lierni Sestorain Saralegui, Tal Schuster, William W. Cohen, Michael Collins, Dipanjan Das, and 3 others. 2022. Attributed question answering: Evaluation and modeling for attributed large language models. ArXiv:2212.08037.
- Arie Cattan, Alon Jacovi, Ori Ram, Jonathan Herzig, Roei Aharoni, Sasha Goldshtein, Eran Ofek, Idan Szpektor, and Avi Caciularu. 2025. DRAGged into conflicts: Detecting and addressing conflicting sources in search-augmented LLMs. In *arXiv preprint arXiv:2506.08500*.
- Shahul Es, Jithin James, Luis Espinosa-Anke, and Steven Schockaert. 2024. RAGAS: Automated evaluation of retrieval augmented generation. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (EACL): System Demonstrations*.
- Claes Fornell and David F. Larcker. 1981. Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research*, 18(1):39–50.
- Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. 2023. Enabling large language models to generate text with citations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. ArXiv:2305.14627.
- Or Honovich, Roei Aharoni, Jonathan Herzig, Hagai Taitelbaum, Doron Kukliansy, Vered Cohen, Thomas Scialom, Idan Szpektor, Avinatan Hassidim, and Yossi Matias. 2022. TRUE: Re-evaluating factual consistency evaluation. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*.
- Kung-Hsiang Huang, Philippe Laban, Alexander R. Fabbri, Prafulla Kumar Choubey, Shafiq Joty, Caiming Xiong, and Chien-Sheng Wu. 2024. Embrace divergence for richer insights: A multi-document summarization benchmark and a case study on summarizing diverse information from news articles. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*. ArXiv:2309.09369.
- Abigail Z. Jacobs and Hanna Wallach. 2021. Measurement and fairness. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*.
- Klaus Krippendorff. 2004. *Content Analysis: An Introduction to Its Methodology*, 2nd edition. Sage Publications.

- J. Richard Landis and Gary G. Koch. 1977. The measurement of observer agreement for categorical data. *Biometrics*, 33(1):159–174.
- Yoonsang Lee, Xi Ye, and Eunsol Choi. 2024. Ambig-Docs: Reasoning across documents on different entities under the same name. In *Proceedings of the First Conference on Language Modeling (COLM)*. ArXiv:2404.12447.
- Nelson F. Liu, Tianyi Zhang, and Percy Liang. 2023. Evaluating verifiability in generative search engines. In *Findings of the Association for Computational Linguistics: EMNLP 2023*.
- Jirui Qi, Gabriele Sarti, Raquel Fernández, and Arianna Bisazza. 2024. Model internals-based answer attribution for trustworthy retrieval-augmented generation. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. ArXiv:2406.13663.
- Hannah Rashkin, Vitaly Nikolaev, Matthew Lamm, Lora Aroyo, Michael Collins, Dipanjan Das, Slav Petrov, Gaurav Singh Tomar, Iulia Turc, and David Reitter. 2023. Measuring attribution in natural language generation models. *Computational Linguistics*, 49(4):777–840.
- Ivan Stelmakh, Yi Luan, Bhuwan Dhingra, and Ming-Wei Chang. 2022. ASQA: Factoid questions meet long-form answers. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. ArXiv:2204.06092.
- Manveer Singh Tamber, Forrest Sheng Bao, Chenyu Xu, Ge Luo, Suleman Kazi, Minseok Bae, Miao-ran Li, Ofer Mendelevitch, Renyi Qu, and Jimmy Lin. 2025. Benchmarking LLM faithfulness in RAG with evolving leaderboards. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP): Industry Track*. ArXiv:2505.04847.
- Han Wang, Archiki Prasad, Elias Stengel-Eskin, and Mohit Bansal. 2025. Retrieval-augmented generation with conflicting evidence. In *arXiv preprint arXiv:2504.13079*.
- Dustin Wright, Sarah Masud, Jared Moore, Srishti Yadav, Maria Antoniak, Peter Ebert Christensen, Chan Young Park, and Isabelle Augenstein. 2025. Epistemic diversity and knowledge collapse in large language models. ArXiv:2510.04226.
- Rongwu Xu, Zehan Qi, Zhijiang Guo, Cunxiang Wang, Hongru Wang, Yue Zhang, and Wei Xu. 2024. Knowledge conflicts for LLMs: A survey. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. ArXiv:2403.08319.